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GNAR-HARX Models for Realised Volatility: Incorporating Exogenous Predictors and Network Effects

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This project introduces the GNAR-HARX model, which combines Generalised Network Autoregressive (GNAR) structure with Heterogeneous Autoregressive (HAR) dynamics and exogenous predictors such as implied volatility. The model is designed for forecasting realised volatility by capturing both temporal persistence and cross-sectional spillovers in financial markets. We apply it to daily realised variance data for ten international stock indices, generating one-step-ahead forecasts in a rolling window over an out-of-sample period of approximately 16 years (2005-2020).
Forecast accuracy is evaluated using the Quasi-Likelihood (QLIKE) loss and mean squared error (MSE), and we compare global, standard, and local variants across different network structures and exogenous specifications. The

best model found by QLIKE is a local GNAR-HAR without exogenous variables, while the lowest MSE is achieved by a standard GNAR-HARX with implied volatility. Fully connected networks consistently outperform dynamically estimated graphical lasso networks.
Overall, local and standard GNAR-HAR(X) models deliver the strongest forecasts, though at the cost of more parameters than the parsimonious global variant, which nevertheless remains competitive. Across all cases, GNAR-HAR(X) models outperform univariate HAR(X) benchmarks, which often require more parameters than the GNAR-based specifications. While the top model found by QLIKE does not use exogenous variables, implied volatility and overnight returns emerge as the most useful predictors when included.

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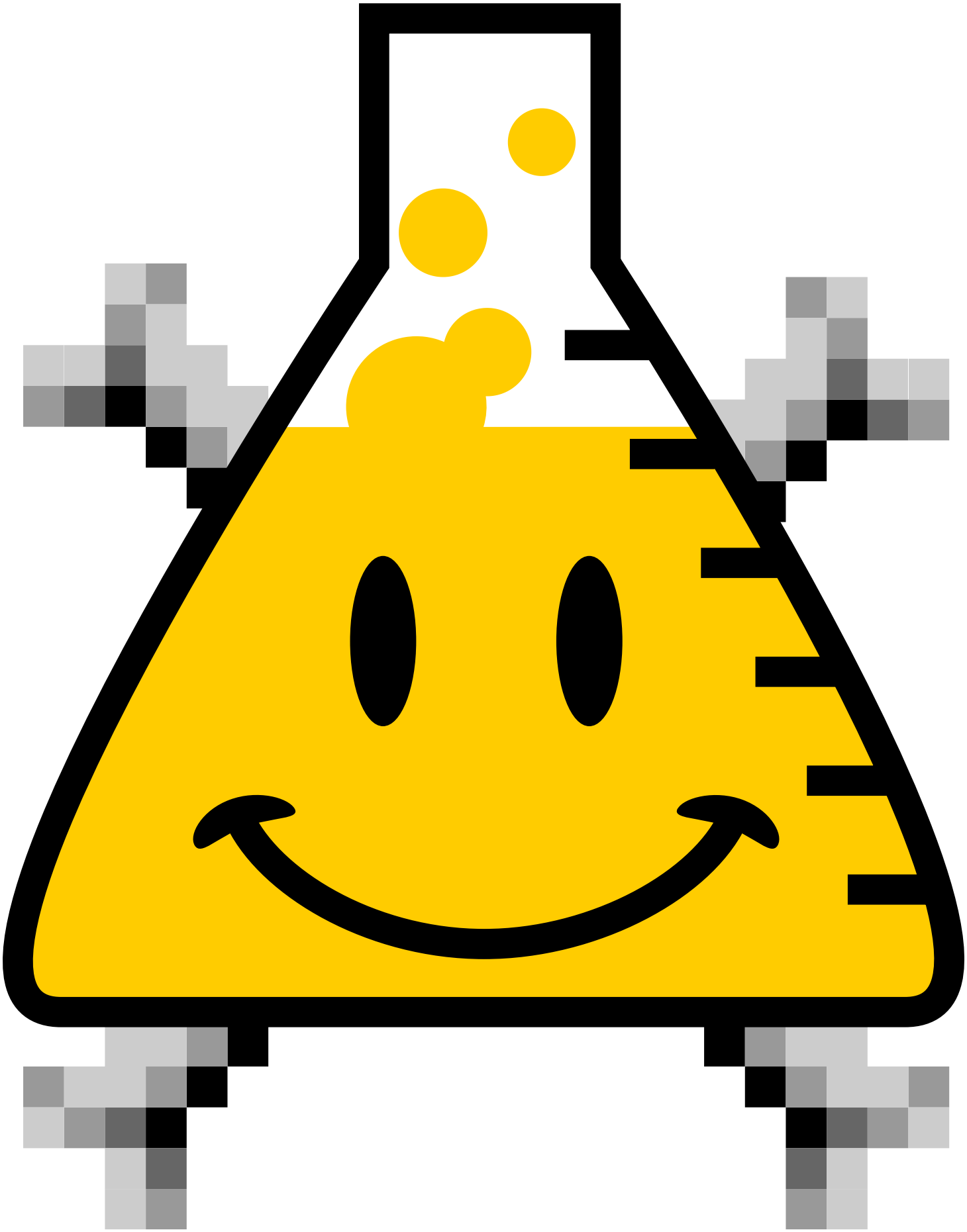

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